

Serial Killer Typology Discovery via K-Means Clustering

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Abstract— This paper presents a cluster analysis based on a data set of 1,531 documented serial killers using the K-Means algorithm to analyze the typologies of serial killers. Following preprocessing, dimensionality reduction via PCA and unsupervised clustering, three typologies were identified, distinguished by decade of activity, active duration, and victim count. The Association rule mining results identified significant co-occurrence relationships between offender characteristics, and the ARIMA time series modeling identified temporal patterns in serial homicide from 1900 to 2022. A Decision Tree classifier trained on cluster labels achieved near-perfect accuracy in reproducing cluster assignments, with the dominant predictive feature being decade. A secondary classification task for predicting the severity category of the victim obtained an accuracy of 77%, with the most important feature being the possible victim count. The results add further to the current burgeoning field of research on the intersection of criminology and machine learning, illustrating the power of unsupervised learning to uncover latent structure in historical offender data.

Keywords: Serial killers, K-Means clustering, PCA, ARIMA, Association rule mining, Decision Tree, criminology, machine learning

I. INTRODUCTION

The study of serial homicide has long been the realm of behavioral psychology and investigative criminology. In the 1970s FBI's Behavioral Science Unit pioneered traditional profiling methods which rely heavily on qualitative clinical assessment. Yet the increasing availability of digitized criminal records has pushed open the door to computational approaches that can surface patterns invisible to manual analysis [18].

Recent work has demonstrated the viability of machine learning for tasks ranging from crime linkage [5, 6] and offender profiling [15, 16] to spatial crime prediction [7, 19, 21]. Clustering methods peculiarly have been applied to identify latent groupings within criminal populations [11], while time series models such as ARIMA have been used to analyze temporal trends in crime rates [3].

In our project we applied clustering and ARIMA to a dataset of over 1500 documented serial killers, with the goal of discovering natural typological sets based on quantifiable behavioral and temporal features. The core research questions are:

- 1) Can K-Means clustering identify meaningful, explainable typologies of serial offenders?
- 2) What features most strongly differentiate between these typological clusters?
- 3) What chronological trends characterize serial homicide activity over the past century?
- 4) What associations exist between offender characteristics such as victim count, active duration, country, and decade of activity?

II. LITERATURE REVIEW

Some research studies have been used to base this work. Research on the consistency of serial homicide behaviour, such as a study by Salfati and Bateman, using the HITS database to find patterns in 69 cases, has revealed that most serial murderers will have regular patterns of behaviour, either instrumental (goal-oriented) or expressive (emotionally driven) [1]. This uniformity facilitates clustering analysis on offenders' data.

Kim et al. [16] were able to predict the motivation of the offender and the number of victims from a 2,500-killer dataset with 81.6% accuracy on predicting the motivation of the offender and RMSE of 4.8 on predicting the number of victims by using the random forest method. The first approach to the class imbalance problem was the MICE technique used to impute missing values, followed by class weights to model the cost-sensitive aspect of the classification problem proposed by Mariani et al. [15], which used ensemble learning techniques (Logistic Regression, SVM, Random Forest, Neural Networks, and Bayesian Networks) to predict nine profile attributes, achieving a balanced accuracy of around 72%.

The statistical analysis of the inter-murder intervals [4] has revealed that power laws are representative of the 'cooling off' period of serial and spree killers, providing no apparent distinction between the two. This work is an extension of the general crime analysis of time series using the ARIMA trend modelling approach [3, 19].

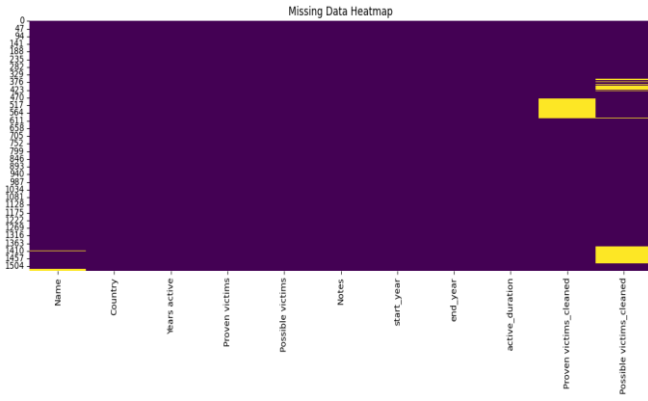
Association rule mining has already been demonstrated to be used in criminological applications of several authors, typically to find co-occurring crime types and/or crime temporal patterns [3, 20, 21]. This is the foundation for an Apriori-based rules application of discretized offender features as done here.

Dakalbab et al. [12] studied the broader field of Artificial Intelligence for crime prediction by reviewing 120 papers, and found that the random forest and Naive Bayes algorithms were the most common. Lima and Karabiyik [10] came up with more specific problems with crime linkage and found four remaining problems to be unsolvable: class imbalance, subjectivity in crime labeling, police data bias, and the limited scope of crime types.

III. METHODOLOGY

A. Dataset Description

The dataset contains 1,531 documented entries for serial killers, with six initial columns: Name, Country, Years Active, Proven Victims, Possible Victims, and Notes. Initially, all columns consisted of object (string) type, which required extensive preprocessing before any analysis could be performed.

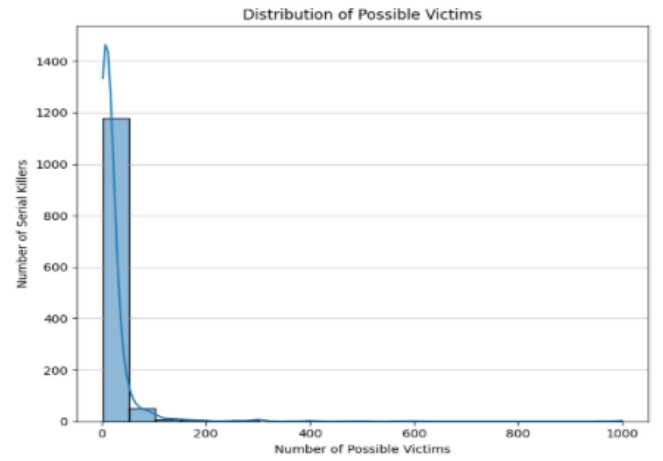
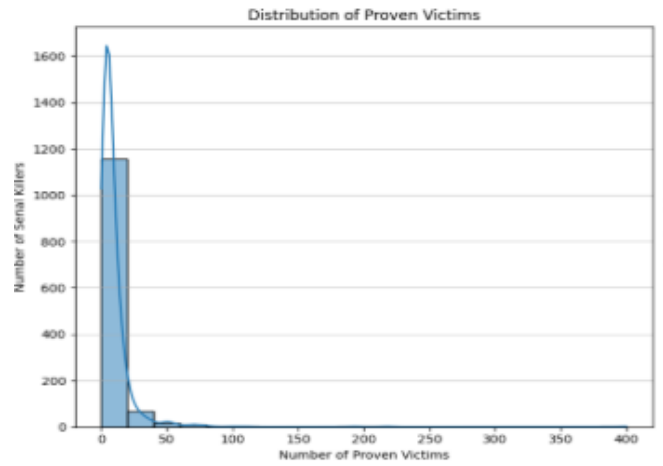
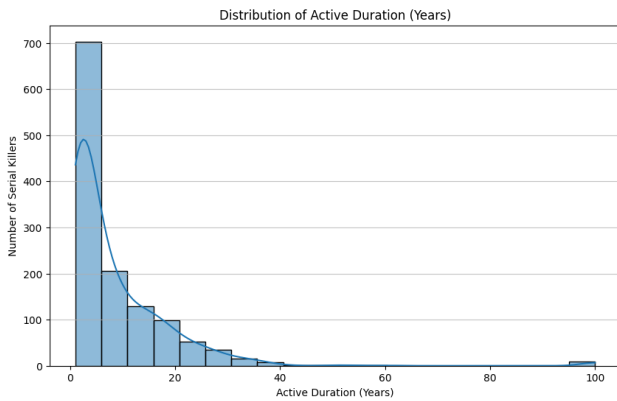


B. Feature Engineering

Three columns, “start_year”, “end_year”, and “active_duration” were extracted by parsing the “Years Active” column. The parsing also handled format variations such as single years, year ranges, approximate years (e.g., “1990s”), and citation-annotated years (e.g., “1992-1999[3]”).

Cleaning the victim count columns required handling numeric ranges (e.g., “193–300+” to 193.0 as the lower bound), plus-annotated values (e.g., 300+ to 300.0), citation-annotated values (e.g., 180[13] to 180.0), and non-numeric values such as “Unconfirmed” which were converted to NaN. After cleaning, 123 proven victim values and 145 possible victim values remained non-numeric and were set to null.

Entries in the Name column that had “unknown” were set to NaN. After dropping all the rows with null values, the dataset was reduced from 1531 to 1258 rows.



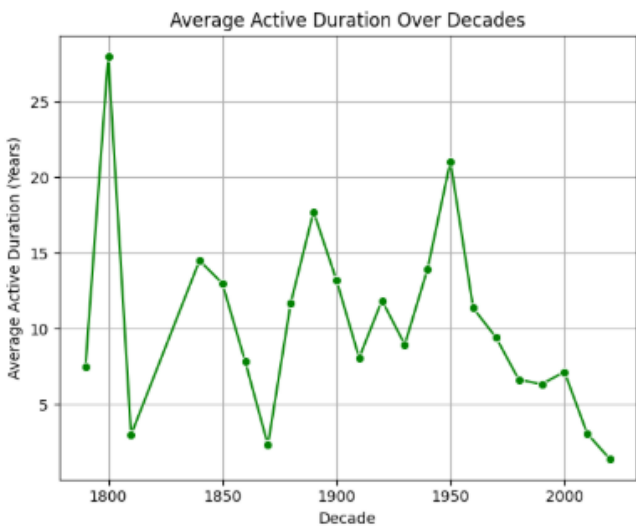
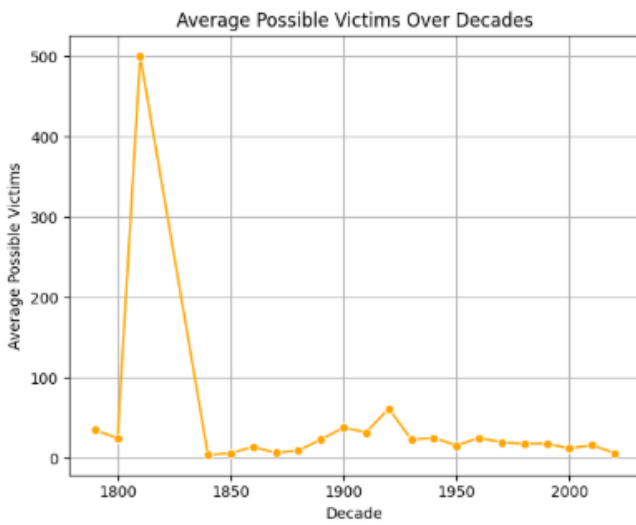
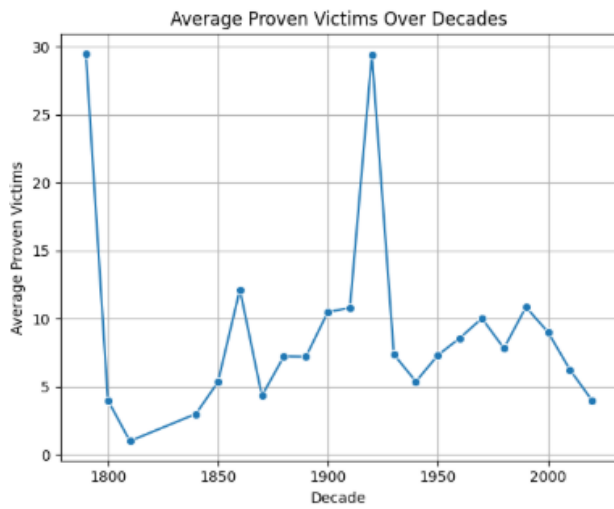
C. Encoding and Scaling

One-hot encoding was used in the Country column, which produced 71 binary country indicator columns. The final dataset for clustering contained 75 columns, including active duration, proven victims cleaned, possible victims cleaned, decade, and 71 country indicators.

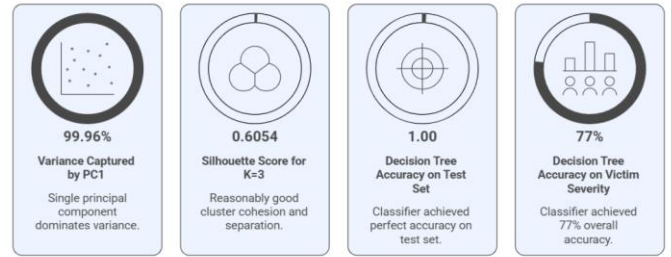
The numeric columns were scaled using MinMaxScaler to bring all values to a range of [0,1]. post-scaling descriptive statistics showed appropriate normalization: means of 0.076, 0.023, and 0.018 for active duration, proven victims, and possible victims, respectively, with a maximum of 1.0 for all three.

D. Decade Analysis

A decade column was created from the “start_year” to analyze the trends. The correlation analysis showed a strong correlation between start_year, end_year, and decade ($r > 0.93$). Active duration showed a moderate negative correlation with start_year ($r = -0.196$), in line with the finding that older records tend to have single years due to insufficient documentation. Proven and possible victims had a moderate positive correlation with each other ($r = 0.547$) but weak correlations with temporal features.



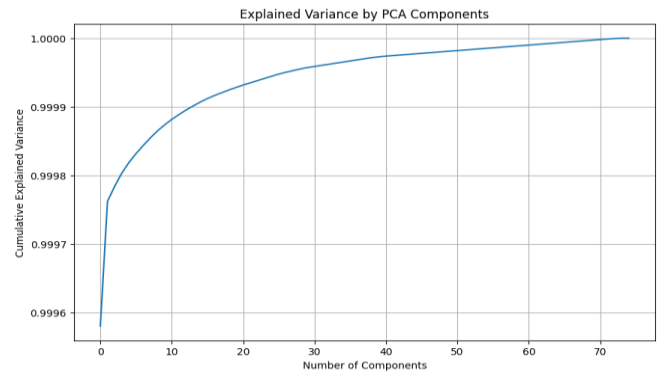
Methodology Overview



PCA and K-Means clustering effectively identified distinct serial killer typologies, with decade being the most significant factor.

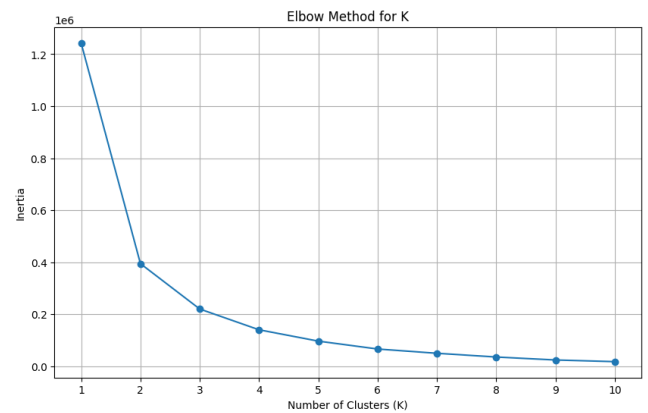
E. Dimensionality Reduction via PCA

The 75-dimensional matrix was used in the Principal Component Analysis. The scree plot showed that a single principal component (PC1) accounted for approximately 99.96% of the total variance. The high concentration of variance in PC1 is due to the dominance of one-hot encoded country features, as they together constitute a high-dimensional but sparse binary matrix. The PCA reduced representation was used for the resulting cluster visualization and violin plot analysis.

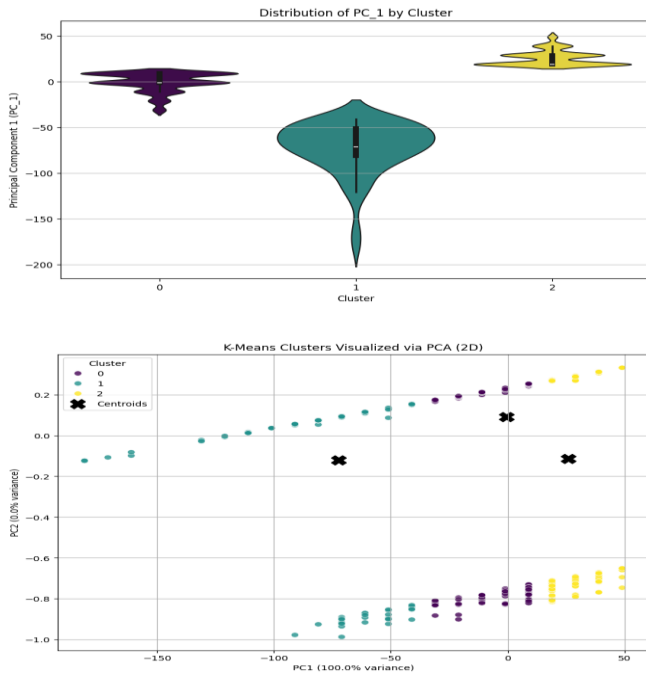


F. K-Means Clustering

The elbow method was applied over $K = 1$ to 10. It evaluated within-cluster sum of squares (inertia). The elbow curve showed the most significant curve at $K = 2$, and then another one at $K = 3$, which we selected to allow richer typological interpretation while also maintaining cluster separation.



K-Means was run with $K = 3$, yielding cluster sizes of 705 (Cluster 0), 410 (Cluster 2), and 143 (Cluster 1). The Silhouette Score for this partition was 0.6054, indicating reasonably good cluster cohesion and separation.



G. Cluster Characterization

Mean feature values per cluster revealed interpretable typological differences:

Feature	Cluster 0	Cluster 1	Cluster 2
active duration (scaled)	0.085	0.103	0.051
Proven victims (scaled)	0.022	0.031	0.024
Possible victims (scaled)	0.018	0.036	0.014
Decade	1970.6	1898.7	1996.9
Count	705	143	410

Cluster 1 centers on the 1900s and is characterized by older offenders, longer active durations, and higher victim counts relative to their era. Cluster 0 centers in the 1970s and actively represents the modal modern offender. Lastly, Cluster 2 centers on the late 1990s/2000s and shows the shortest active durations which would be consistent with improved law enforcement detection decreasing active timeframes.

Top Countries for Each Cluster:

Countries	Cluster 0	Cluster 1	Cluster 2
United States	607	113	251
Soviet Union	17	2	-
France	7	5	-
Russia	-	-	27
Germany	-	4	-

Top countries per cluster showed us U.S. dominance across all three clusters reflecting dataset composition rather than genuine geographic typology. Notable secondary contributors include Soviet Union records in Cluster 0, Russia in Cluster 2, and France and Germany in Cluster 1.

H. Anova Validation

One-way ANOVA tests confirmed that cluster differences were statistically significant for active duration ($F = 16.36$, $p < 0.0001$), possible victims ($F = 8.64$, $p = 0.0002$), and decade ($F = 2917.22$, $p < 0.0001$). Proven victims did not reach significance ($F = 2.75$, $p = 0.064$), suggesting victim count confirmation is noisier than temporal and duration features.

ANOVA Test Result

Feature	active_duration_original	Proven victims_cleaned_original	Possible victims_cleaned_original	decade
F-statistic	16.3625	2.7542	8.6429	2917.2210
P-value	0.0000	0.0640	0.0002	0.0000
Interpretation	Statistically significant difference	No statistically significant difference	Statistically significant difference	Statistically significant difference

I. Association Rule Mining

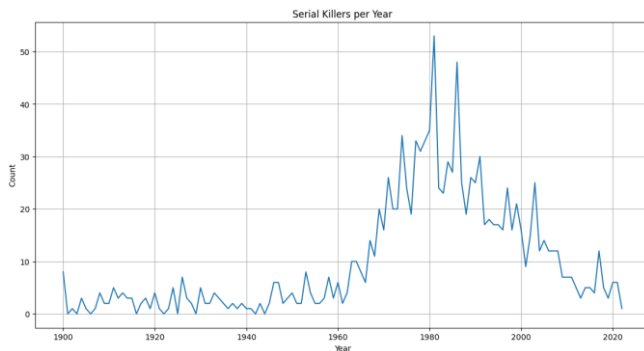
The dataset was categorized into binary features for Association Rule Mining (ARM) using the Apriori algorithm. Victim counts were stored into categories (1–5, 5–10, 10–20, 20–50, >50) and decade was also binned into historical periods. The resulting ARM DataFrame contained 96 binary columns across 1,258 rows.

Top rules by lift identified strong co-occurrence between victim ranges: offenders with >50 possible victims almost always have >50 proven victims (confidence = 1.0, lift = 14.13). Rules by confidence revealed that offenders with 1–5 possible victims in any decade consistently have 1–5 proven victims (confidence = 1.0), and that a majority of 1970s–1980s offenders with low active durations have U.S. country records (confidence = 1.0, lift = 1.30).

J. Time Series Analysis (ARIMA)

A yearly series of serial killer activity published between 1900 and 2022, shows activity peaking at an average of 55 per year around 1980's followed by a sharp drop thereafter.

The series shows non-stationarity according to an Augmented Dickey-Fuller test statistic (-1.60), so an ARIMA(1,1,1) model has been estimated after differencing the data. Results indicate that the moving average (MA) component was important (coef = -0.582; $p < 0.001$), while the autoregressive coefficient (AR) was not important (AR coef; $p = 0.515$). Trend analyses reconfirmed that serial killers showed a mean activity difference between years before and after 1970 (prior mean = 3.49 murders per year; post mean = 18.28 murders/year). Since 2000, the mean number of serial murders per year has decreased to 8.83, a continuing downward trend since reaching an all-time high in the early to mid 1980's.



K. Decision Tree Classification

A Decision Tree classifier was built using the full 75-feature dataset to determine cluster interpretability as well as the learnability of the cluster structure. Using an 80/20 train-test set of 378 samples, the classifier achieved perfect accuracy (1.00), with precision, recall, and F1 scores all equal to 1.00 across the 3 clusters.

Feature importance analysis demonstrated that the decade alone accounted for 100% of the decision tree's information gain, while all other features gained 0%, confirming that the K-Means partitioning was used for the PCA compressed space ruled by temporal variance created decade-based clusters.

Another Decision Tree was used for building a victim severity classification (low/medium/high/extreme based on victim count), and had an overall accuracy of 77%. The most important features were Total Possible Victims (0.591), length of time active (0.196) and Decade (0.117). The country indicators weren't important feature predictors.

Classification Report:				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	212
1	1.00	1.00	1.00	43
2	1.00	1.00	1.00	123
accuracy			1.00	378
macro avg	1.00	1.00	1.00	378
weighted avg	1.00	1.00	1.00	378

ADF Test Result:
ADF Statistic: -1.5987
p-value: 0.4843
Critical Values:
1% : -3.4891
5% : -2.8872
10% : -2.5805

IV. RESULTS AND DISCUSSION

A. Cluster Typologies

The three clusters map onto explainable documented typologies:

- **Cluster 1** — Historical Era (Early 1900s): The smallest cluster of 143 is characterized by older case records, longer apparent active durations (due to poor historical documentation, we have few facts), and higher possible victim estimates relative to proven victims, showcasing the limited forensic capabilities of the era. This cluster also captured offenders like European poisoners and serial killers, where the victim hallmarks were very uncertain.
- **Cluster 0** — Mid-to-Late 20th Century (1960s–1980s): The largest cluster of 705 is at the center of the peak era of documented serial homicide. It exhibits moderate active durations and victim counts. An overwhelming number of U.S. cases in this cluster reflects both actual homicide activity as well as better chronicle during this period. It also aligns with behavioral profiles rigorously studied in FBI criminological literature.
- **Cluster 2** — Contemporary Era (1990s–2010s): The runner-up cluster with 410 is categorized with the shortest active durations, consistent with the hypothesis that improved forensic techniques, DNA databases, and law enforcement strategies enable faster arrests. Lower possible vs. proven victim ratios showed that modern investigations have better evidentiary standards.

B. Limitations

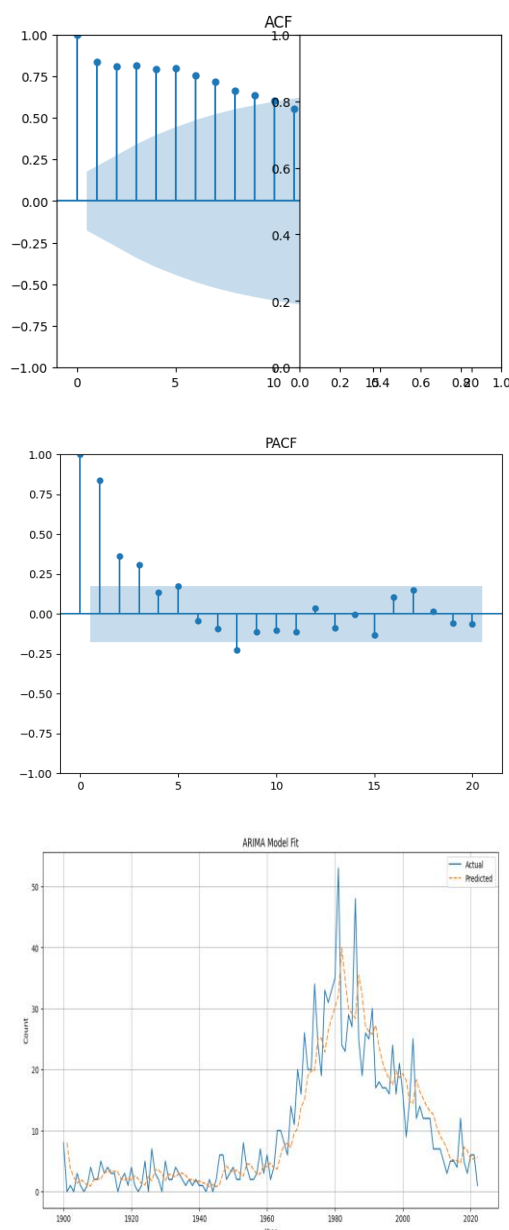
The Decision Tree analysis shows that decade had the strongest influence on the clusters, meaning the K-Means model mainly grouped cases by time period rather than behavioural patterns. This likely happened because PCA reduced the 75 features into one main component, which was heavily affected by country encoding and temporal features.

To produce more behaviour-based clusters, other methods could be explored, such as clustering only behavioural variables like victim count and active duration, removing country encoding, using DBSCAN or Gaussian Mixture Models, or applying NLP techniques to the Notes column to identify patterns in offender behaviour.

The dataset may also contain selection and reporting bias, since it mainly includes well-documented convicted offenders from countries with accessible records. The large proportion of U.S. cases in Cluster 0 (77%) may reflect better record availability rather than actual prevalence.

C. Time Series Findings

The ARIMA model describes the overall trends of data showing increased rates of activity between 1970-1980 and decreased rates afterwards. However, the high Jarque-Bera (92.81) and heteroskedasticity (H-Test $P < 0.001$) of the residuals suggest that they aren't normally distributed. The unstable yearly counts as well as the short time span after 1950 limit the long-term predictive ability of the model. The primary purpose of the model is to provide a descriptive record of statistically significant increases and decreases in documented serial homicide activity over the course of the 20th Century.



IV. CONCLUSION

This study applied K-Means clustering, PCA, ARIMA time series analysis, association rule mining, and Decision Tree classification to a historical serial killer dataset. There were three distinct temporal

typologies found (Silhouette Score 0.6054, validated through ANOVA) that were completely reproducible with a decision tree classifier using only the decade as a feature to separate them.

Association rule mining showed there were many interpretable patterns discovered concerning when killers were operating in different countries around the same time and to what level of victimization occurred. ARIMA time series predictions showed a rapid increase in reports of violence against children until 1970 at an average case rate of 3.49 per year and again from 1980 to 2000 with 18.28 per year.

For future research, it would be wise to focus on extracting more detailed behavioral features from textual fields, as well as trying alternative clustering methods that are less susceptible to the impact of time on results. Finally, it is important to explore how to combine unsupervised typology creation with supervised prediction models across multiple digital forensics data sources as discussed in the literature offered by [17]. Unsupervised typologies when combined with supervised prediction models shows that the potential of utilizing computational techniques in criminology can be valuable understandings.

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